

NEXUS CAMPUS

An Augmented education app

DONE BY,
AMRITHA A
SNEHA S



PROJECT OVERVIEW

Nexus Campus is an Augmented Reality (AR) based educational application created for students from Grade 3 to Grade 8, with the current prototype focusing on Grade 8 learners. The aim of Nexus Campus is to make textbook learning more interactive, engaging, and easier to understand using AR visualizations. The app allows students to scan textbook images or keywords to view 3D AR models of related topics such as the solar system, plant cell, or chemical elements directly on their mobile devices. It also includes a chatbot assistant named Nexi, which helps explain topics, answer questions, and guide students through lessons. At present, the Figma design prototype has been completed, showing the user flow that includes grade selection, student dashboard, camera integration for scanning, textbook browsing, and chatbot interaction. The project is currently in the research and feature development stage, focusing on identifying creative and practical features that can be implemented in the upcoming AR prototype.



Additional Features



As part of the research phase, several new features are proposed to make Nexus Campus more interactive and engaging for students. These include interactive AR labels that show names and details when tapped, a textbook snapshot window to display related content, a vocabulary booster for key terms, cross-subject concept linking to connect related topics, voice command support for easier interaction, and a scroll-up panel to show summaries or fun facts. These features aim to make learning more visual, connected, and enjoyable for students.



Interactive AR Labels

Students can tap on specific parts of an AR model to view interactive labels and short descriptions. They can scale, rotate, and move the model to observe it from all sides for a better understanding. For example, when viewing the solar system, students can zoom in on a planet or rotate it to explore details. This feature encourages active learning and helps visualize complex topics easily. It makes learning more engaging, interactive, and conceptually clear.



Textbook Snapshot Window

While viewing the AR model, a small window displays the related textbook image or key definition for quick reference. This helps students connect what they see in AR with the textbook content. The feature makes learning smoother by combining visuals and text in one view. It supports better understanding and memory retention. This simple addition enhances clarity and learning efficiency.



Word Explorer

The chatbot Nexi can introduce 3–5 important words related to the current topic, along with their meanings and simple examples. This helps students understand key terms while learning new concepts. It builds vocabulary naturally through the lesson content. The feature also supports better reading comprehension and memory. Overall, it makes learning smoother and more meaningful for students.



Concept Connector

This feature links related ideas across different subjects to help students see how concepts are connected. For example, a topic like “Cell Division” in Science can relate to “Growth Patterns” in Math. It encourages broader thinking and helps students understand real-world applications. By connecting subjects, learning becomes more integrated and meaningful. This promotes curiosity and deeper understanding beyond textbooks.



Voice Command Integration

This feature allows students to talk to Nexi using simple voice commands for quick responses. They can say phrases like “Show me the model” or “Summarize this topic” instead of typing. It makes the learning experience faster and more natural for young learners. The feature also supports accessibility for students who prefer speaking over typing. Overall, it adds convenience and makes interaction more engaging.



Information Slider

This feature adds a small sliding panel below the AR model that shows extra information like summaries, fun facts, or key points. Students can swipe up to read more or swipe down to hide it. It keeps the screen clean while still providing useful details. The feature encourages curiosity and deeper understanding of the topic. It makes learning more organized and easy to follow.



CONCLUSION

Nexus Campus is designed to make learning more engaging and interactive using AR technology. The proposed features aim to simplify difficult topics and create a fun learning experience. By combining visuals, interaction, and accessibility, the app supports better understanding for students. The next step is to develop and test the working prototype. Overall, it has strong potential to make learning more meaningful and enjoyable.





**Thank
You!**